

KindaRails2Shell: how a MATLAB file reads your secrets and pops a shell on Rails (Rails technical details)

KindaRails2Shell: how a MATLAB file reads your secrets and pops a shell on Rails (Rails technical details) - Author not stated, GitHub.

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Content

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This reference is a source-code repository. The archive preserves its documentation at an exact commit; the code itself stays in a private mirror and is never checked out, built or run.

- Repository: <<https://github.com/rails/rails-forensics-CVE-2026-66066>>
- Commit: 81805d617de6b82c35908ff11bf7df6e5e2e6dba
- Documents preserved: 2

LICENSE

Blob [d03d66cf4600](#), 1088 bytes, at commit [81805d617de6](#).

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README.md

Blob bf9463740184, 8030 bytes, at commit 81805d617de6.

rails-forensics-CVE-2026-66066

Tooling and method for answering two questions about a Rails application and CVE-2026-66066, the Active Storage and libvips arbitrary file read known as KindaRails2Shell:

1. **Was I vulnerable, and for how long?**
2. **Was I exploited, and if so, what left the building?**

The advisory [GHSA-xr9x-r78c-5hrm](#) explained the vulnerability and named the patched versions, but did not disclose enough detail to evaluate these questions. This project is intended to help with that.

This is a documentation and skill repository rather than a library. Nothing here is installed into your application. An agent reads it and works against your application checked out somewhere else, and the one script that runs in production is a single self-contained file you copy into a container.

The vulnerability in one paragraph

Rails decides a blob may be image-processed by reading a `content_type` column that a direct upload lets the client set without the bytes ever being examined. libvips then decides what the file actually is by sniffing magic bytes. A file whose first ten bytes claim `MATLAB 5.0` is routed to libvips' MATLAB loader, which hands it to libmatio, which dispatches on a *different* byte range and finds MAT 7.3, which is HDF5. HDF5's External File List lets a dataset's bytes live in another file named by path and offset, so rendering the "image" reads an attacker-chosen file off the server and returns its contents as pixels. The same confusion, twice, at two layers that cannot see each other's fields.

`reference/the-attack.md` traces the whole call chain, from the inbound request to the point where libmatio opens the target file, with the four load-bearing junctions marked. `reference/the-investigation.md` then explains why the attack is investigable at all: it leaves three artifacts of increasing strength, and the strongest of them is the rendered variant, which holds the stolen bytes as pixel values in your own object store.

Start here

The two skills run in order. The first produces the exposure window; the second cannot start without it.

Skill	Question	Produces
kr2s-was-i-vulnerable	Was this application ever exposed?	An exposure window and the facts the sweep needs
kr2s-was-i-exploited	Did anyone use it, and what did they get?	A written forensic analysis

Both are written for an agent, and both are readable by a person who wants to do the work by hand. Each has a `SKILL.md` entry point and a `references/guide.md` holding the actual process, its eval checks and its failure modes.

If you are patched and want to know whether you should care, run the first one. Its deliverable is a window rather than a yes or no, because for most people reading this the present-tense answer is "no, we patched," and the useful answer is which years an investigation would have to cover.

Getting them into your agent

Clone this repository and keep the clone. The two skills share the reference documents in `reference/`, and their guides read those by repository-relative path, so a skill directory copied somewhere on its own will dead-end.

The easiest thing is to start your agent inside the clone and tell it where your application is. `AGENTS.md` orients it from there, including the part people get wrong, which is that this repository is the tool and not the subject.

Otherwise, point at the file and say so, which works in any agent that can read local files:

```
Read /path/to/rails-forensics-CVE-2026-66066/skills/kr2s-was-i-vulnerable/SKILL.md
and follow it. My application is at /path/to/my-app.
```

In Claude Code you can instead install them so they trigger on their own, by symlinking rather than copying, which keeps them pointed at the clone:

```
ln -s "$PWD/skills/kr2s-was-i-vulnerable" ~/.claude/skills/
ln -s "$PWD/skills/kr2s-was-i-exploited" ~/.claude/skills/
```

Use `.claude/skills/` inside a project instead of `~/.claude/skills/` if you would rather scope them to one application.

Either way, tell the agent where your application is. The skills are explicit that this repository and the application under investigation are two different roots, and that they may read the second but never write to it.

What is in here

AGENTS.md	orientation for an agent started in this directory
reference/the-attack.md	how the vulnerability works, traced call by call
reference/the-investigation.md	why an investigation is possible, and where it runs out
skills/	the two skills
RUNBOOK.md	operator's guide to the scanner: flags, verdicts, resuming
lib/crafted_mat_file.rb	the detector
bin/	the scanner and three analysis utilities
test/	the test suite

The two reference documents are the plain-English half of this repository and they stand on their own. Read them if you want to understand the problem, run the skills if you want an answer about a particular application.

`bin/kr2s_scan_active_storage_blobs.rb` is generated from `lib/` by `rake build` and committed, so running it in production needs no build step and no second file.

The detector

`lib/crafted_mat_file.rb` identifies the crafted file this attack requires. It reads two header fields out of the first 128 bytes of an object and needs nothing else, which is why the scan can classify a candidate from a ranged read rather than downloading it. `reference/the-attack.md` explains which fields and why no legitimate writer produces that combination.

No payload generator

This repository deliberately ships no code that builds crafted files, and no crafted files. The detector's tests synthesize the 128 bytes of header they need in pure Ruby, which is all the detector reads. `reference/the-attack.md` describes the structure completely, so nothing is being withheld from a reader trying to understand the attack. What is withheld is a working generator that could be retargeted at an arbitrary path by changing one argument.

Scope

Covered: Active Storage, on applications using the `:vips` variant processor, where variant records were tracked.

Not covered: any other place an application hands user-supplied files to `libvips`. Other upload libraries, direct object-store clients, and bespoke avatar or logo systems each need their own sweep, and the scanner here will not do it. `kr2s-was-i-vulnerable` will make you enumerate them so they are named rather than silently omitted from a conclusion.

A clean result from this tooling is strong evidence, not proof. Both skills say so, repeatedly and specifically, and the analysis template has a limits section that is not optional.

Provenance

This came out of a real investigation at 37signals, sweeping a large production Active Storage store over the full period the vulnerability existed. The tooling correctly identified the test files uploaded by the security researchers who reported the flaw, which is the only external evidence available that the detector finds what it is supposed to find.

The worked analysis in `kr2s-was-i-exploited/references/analysis-template.md` is fictional. It is modelled on the real one, but every identifier, address and date in it is invented.

References

- [GHSA-xr9x-r78c-5hrm](#), the Rails security advisory
- [CVE-2026-66066](#)
- [Ethiack's write-up](#)
- [RyotaK's write-up](#)

Take the patched Rails version numbers from the advisory rather than from anything written here.

Development

```
mise exec -- rake    # build and test
```

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Archived from <https://github.com/rails/rails-forensics-CVE-2026-66066>. Rendered offline from the local Markdown copy.